CM4RM

Scientific reports and presentations

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Understand and interpret your simulation results

Check

1. Data check: metfiles , soil files , management → does it makes sense? Plot your data

2. Set up your base simulation

3. Check your base simulation → does it makes sense (simulation vs real world, agronomic knowledge) ? Plot your data → If not go back 2 or even 1

Understand and interpret your simulation results

Understand

1. What are your output variables (biomass, lai , yield etc) , axes , scales etc
2. Simulation vs real world observation
3. How does the model simulate real world problems?
4. What does the model do and why ?
5. Literature back up !

Recommendation:

Read 1 or 2 modelling studies for inspiration

Scientific report

- Title page (your names, Matr.Numbers, Title , Submission date , Name of Examiner)
- Table of contents
- Table of Figures and Tables
- Abstract (max 300 words)

Scientific report

- Introduction
  - Research gap
  - Relevance
  - Research question / hypothesis
  -
- Material and Methods
  - Describe the situation you are simulating
  - Which data did you use – data sources etc
  - Describe relevant details of your model set up
  - Provide all information necessary to recreate your simulation

Scientific report

- Results
  - DESCRIBE your simulation results
  - No interpretation yet

- Discussion
  - Analysis and Interpretation of your study results
  - Application -> what does this mean for the real world (remember your study question)
  - Shortcomings / limitations of your study
  - Future prospects

- Conclusion
  - Draw conclusions don’t repeat everything from the results or discussion !!

Scientific report

- Max 10 pages – text (from Introduction conclusion !!!!) Figures will not be counted
- Margins – left , right and top 0.98 inches , bottom 0.78
- Font: Arial – 11 pt for the text , 12pt for headlines
- Paragraph – “justify”
- Page numbers
- Line spacing = 1.5
- Figures MUST have captions (9 pt, line spacing = 1) → include all necessary information , legends , colors, titles etc
- Tables MUST have headers (9 pt, line spacing = 1)

Overall form, structure and language approx 32 points
Cover page
Table of contents
Language (general, and correct use of technical terms and scientific terminoolgy)
Citations/references (to literature, figures etc)
Grammar, typos, and spelling
Reference list
Length (within in given page limit)
overall form (layout, page numbers line distance etc
overall structure follows scientific paper standard
Meaningful structure within the separate sections

Abstract 10
within word limit
Good coverage / summary of the indiidual parts of the paper
Additional Notes:

Introduction 12
Justification of the relevance of the topic , presentation of the current status quo in research and the current knowledge gap
Derviation, development and Formulation of the hypothesis/research question
Additional Notes:

Additional Notes:

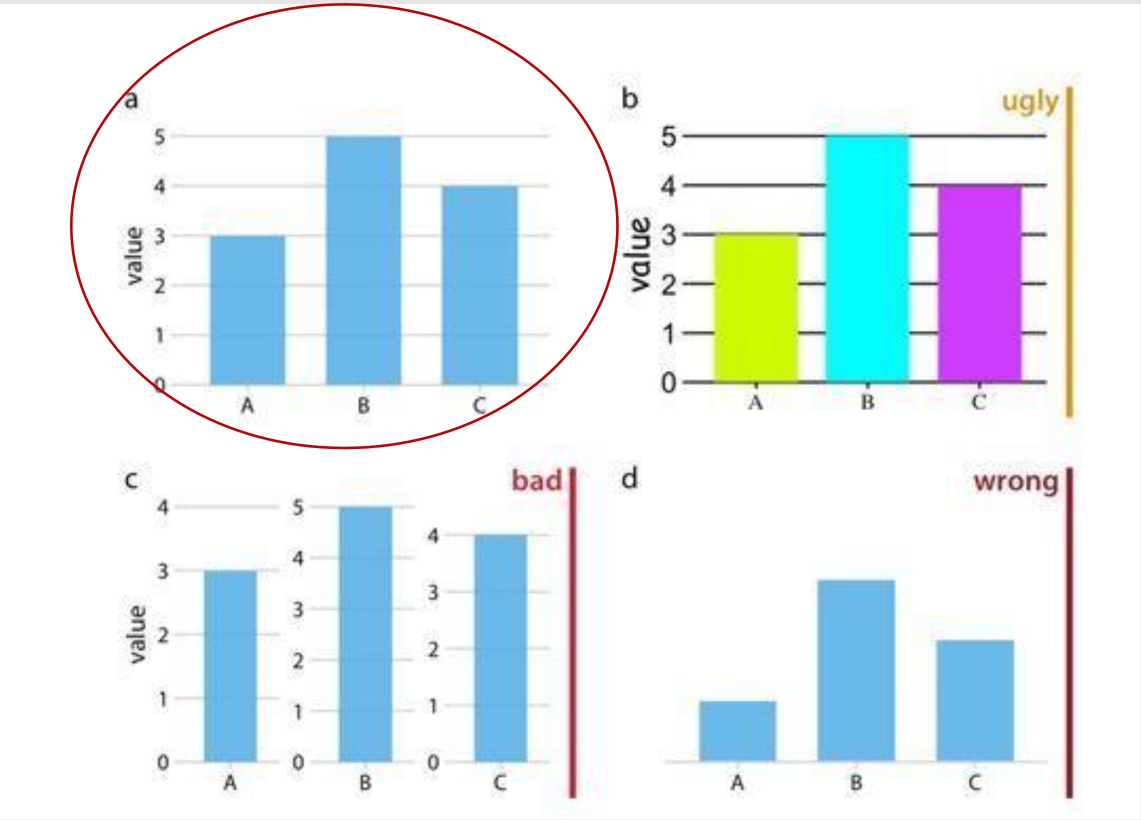
Methods	10
use of appropriate methods	
Descriptoin of all important methods (measurement methods, model set up etc) so that one could reproduce the expriments/simulations	
Presentation of relevant formulas and calculations	
appropriate level of Independent Working	
Additional Notes:	
Results	25
All important results are described, concisely, correctly and with appropriate detail; without contradicting the figures	
Data is visualized appropriately following the general rules of data visualization)	
Additional Notes:	
Discussion	46
Meaningful, justifiable , comprehensible interperations of own results	
Thought process (logic, stringency)	
Breadth and depth of topic how the topic is covered/handled; Own results are discussed in the context of currrent research	
Critical evaluation of this study (limitations etc) points: 5	
Presentaion of wider implactions/ practical applications of results / further research questions	
Additional Notes:	
Conclusion	10
brief Summary of key outcomes and wider implications/ consequences (not just a repetition of all results)	
Presentaion of follow up research questions	
Additional Notes:	

Scientific presentations

- Background
  - Research gap
  - Relevance
  - Reseasrch question / hypothesis
- Material and Methods
- Results
- Discussion
- Conclusion

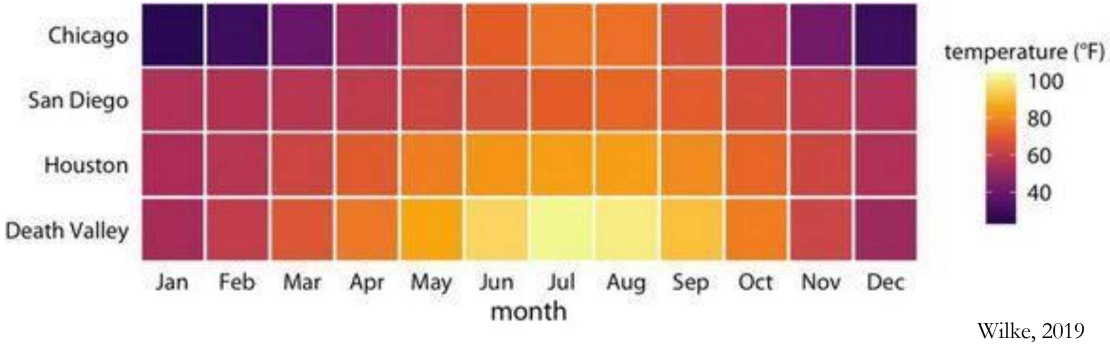
  - Focus on the most important aspects !! Don't present everything
  - 15 min presentation
  - 10 min disucssino / questions

Principals of scientific data visualization

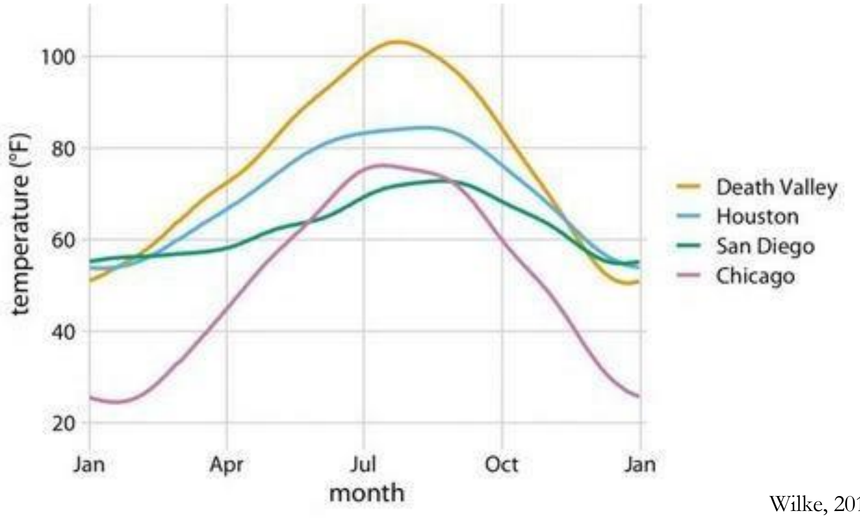


Wilke, 2019

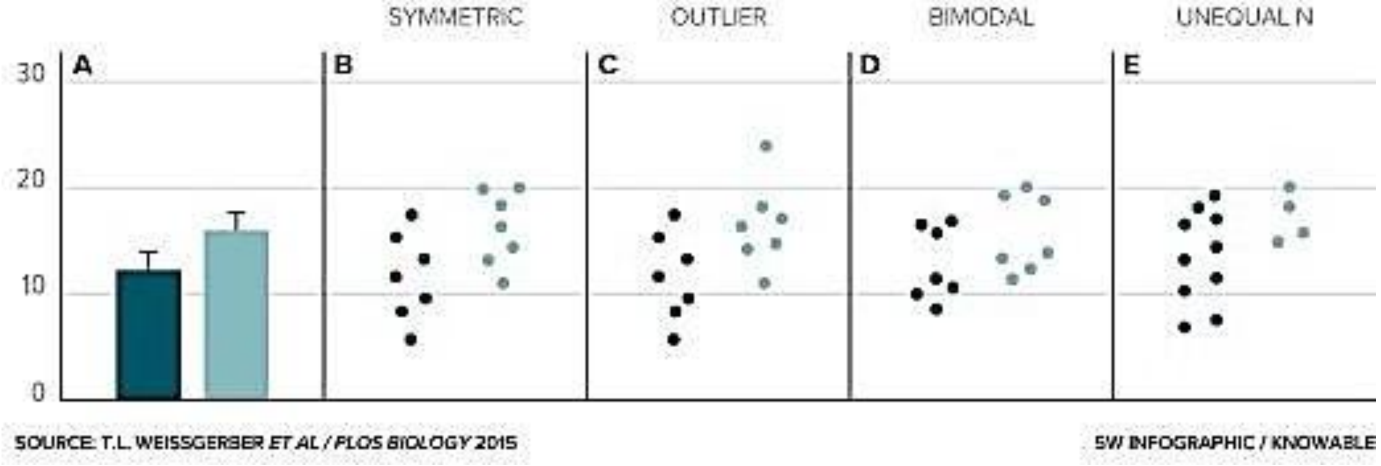
Pick the right plot type



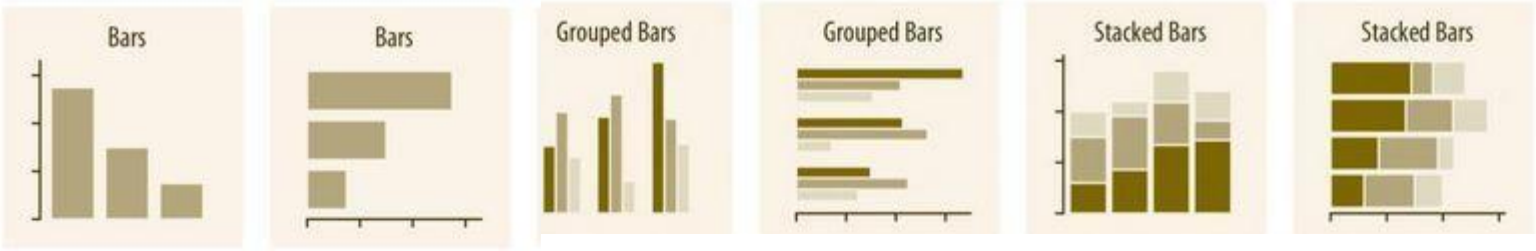
Wilke, 2019



Wilke, 2019



- Amounts

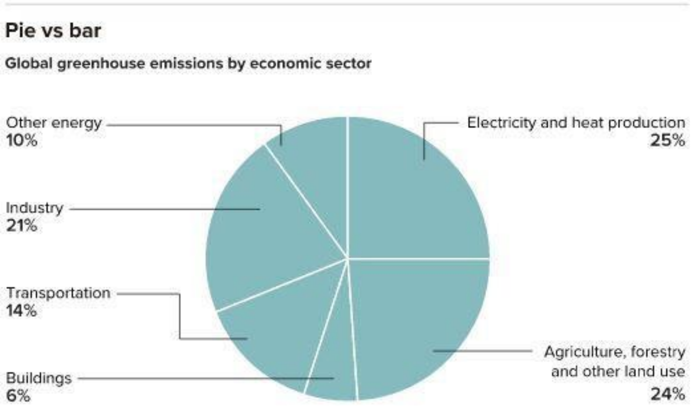


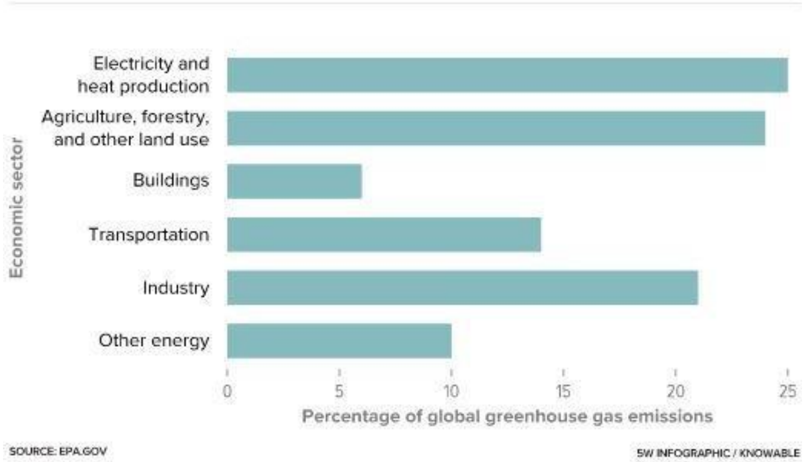
for two or more sets of categories

- Proportions

not recommended

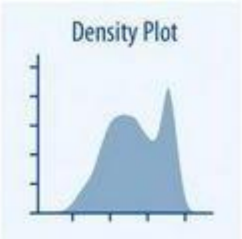
better

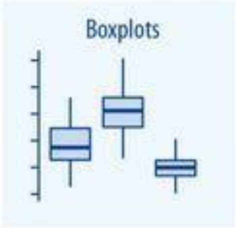




Wilke, 2019

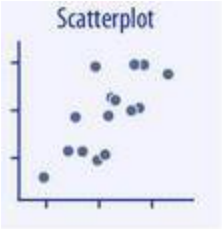
- Distribution



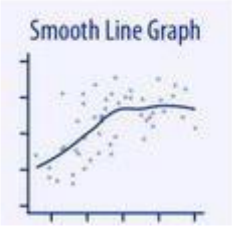


for overall shifts among the distribution

- X – Y relation

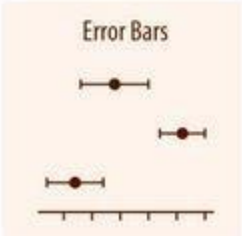


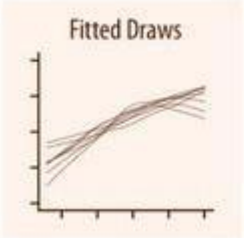
one variable
relative to another



x axis represents time or treatments

- Uncertainty





indicate the range of likely values

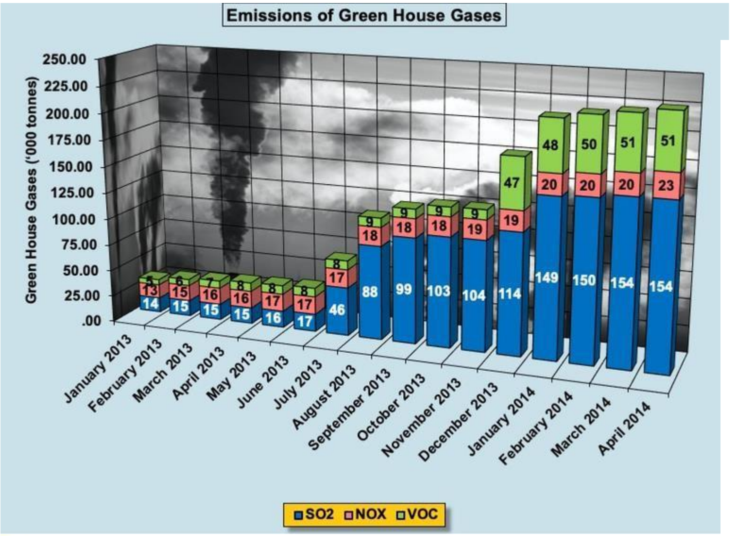
Wilke, 2019

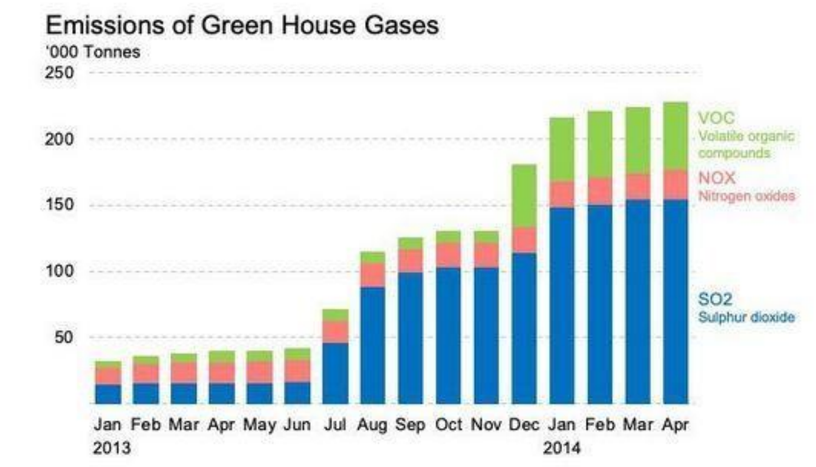
- data visualization should make complex information **easily accessible** to the reader

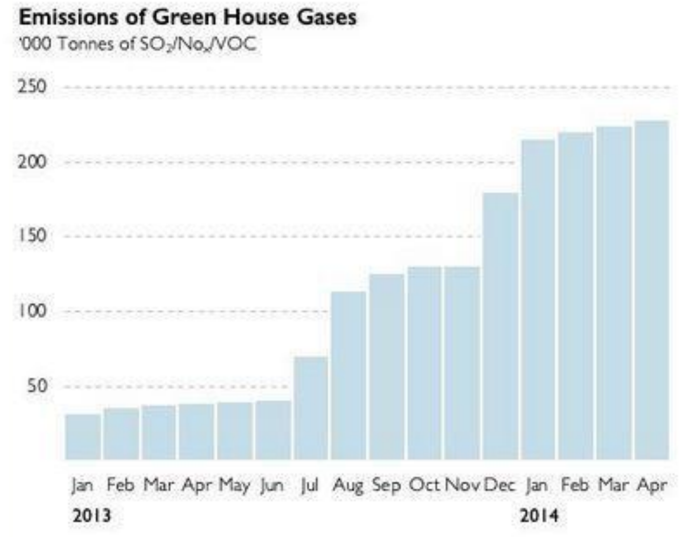
- **quickly** and **effectively** convey scientific information

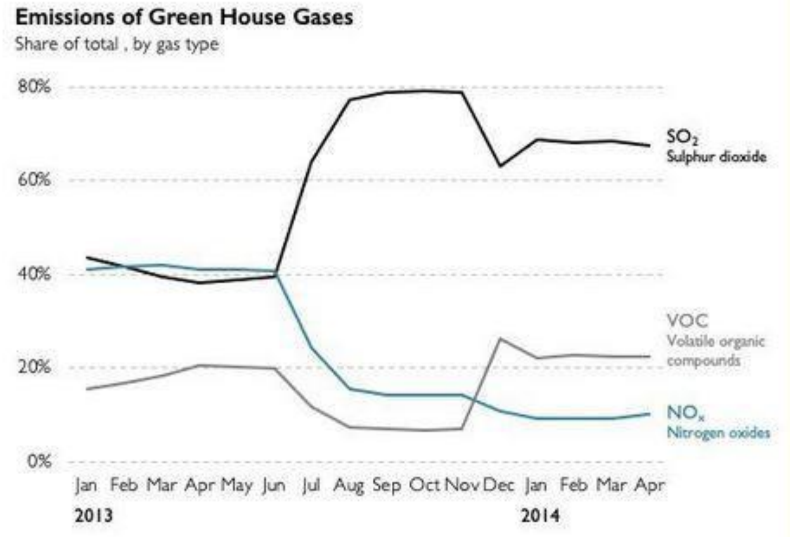
- → every visual element has to serve a purpose

- plots have to be self explanatory

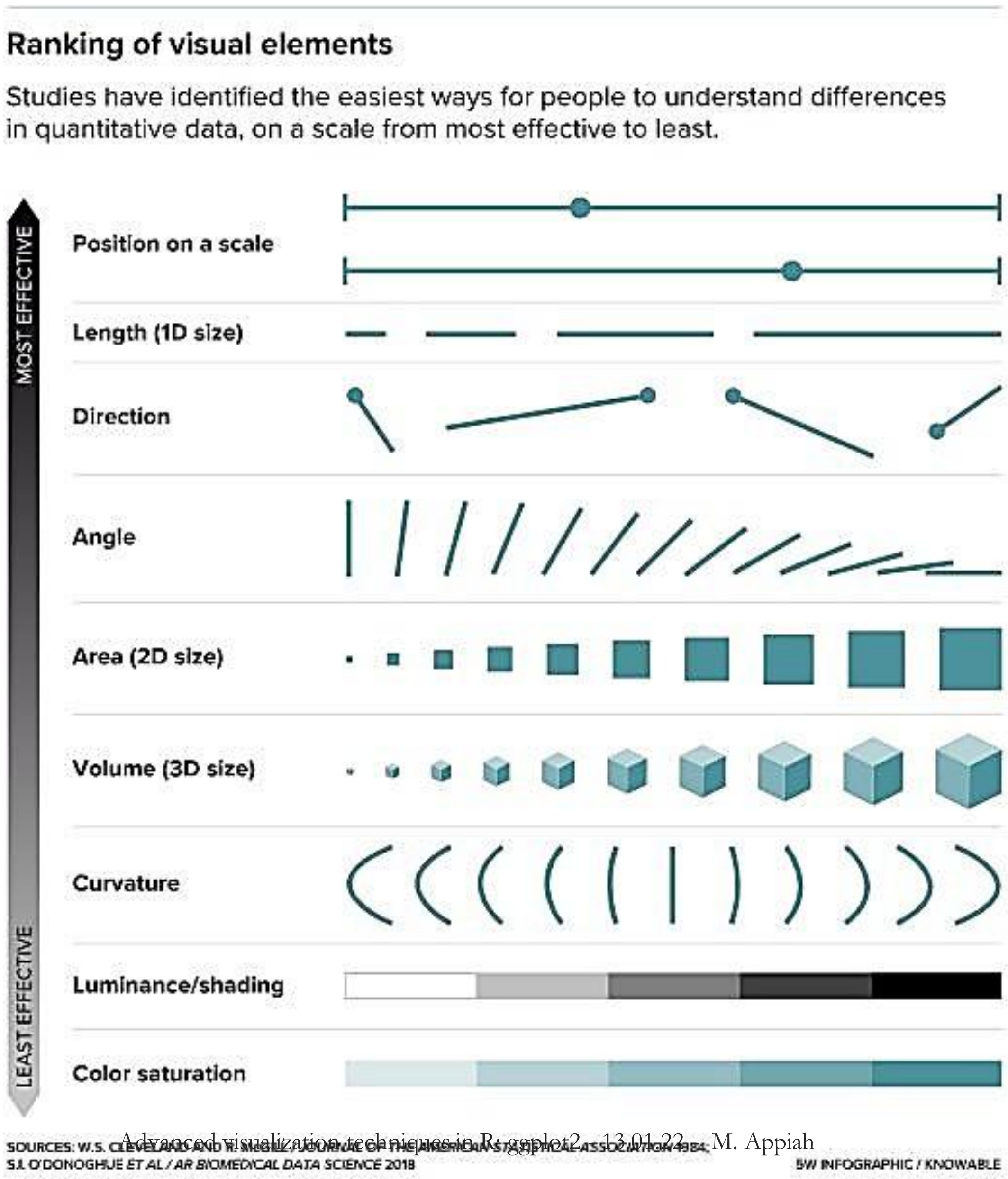








https://simplexct.c$_1$o$_9$m/data-ink-ratio



Choose your visual element (geometries) carefully!

Colors

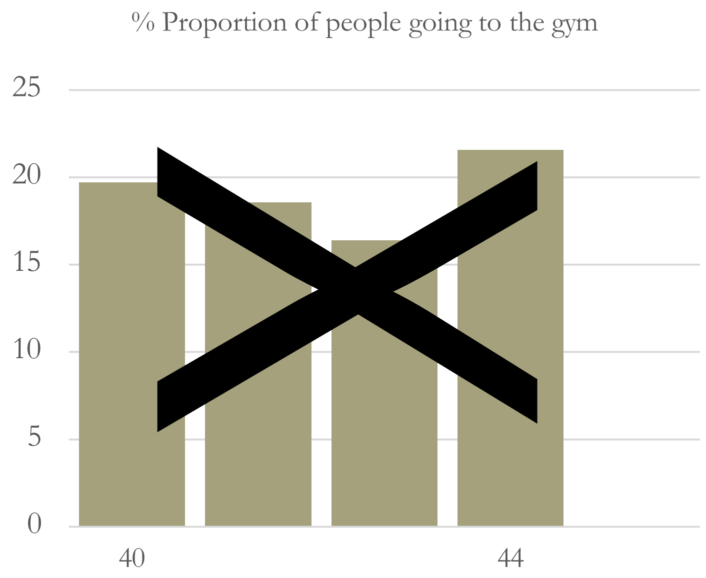
- colors can draw attention to important information
  but can also mislead the reader

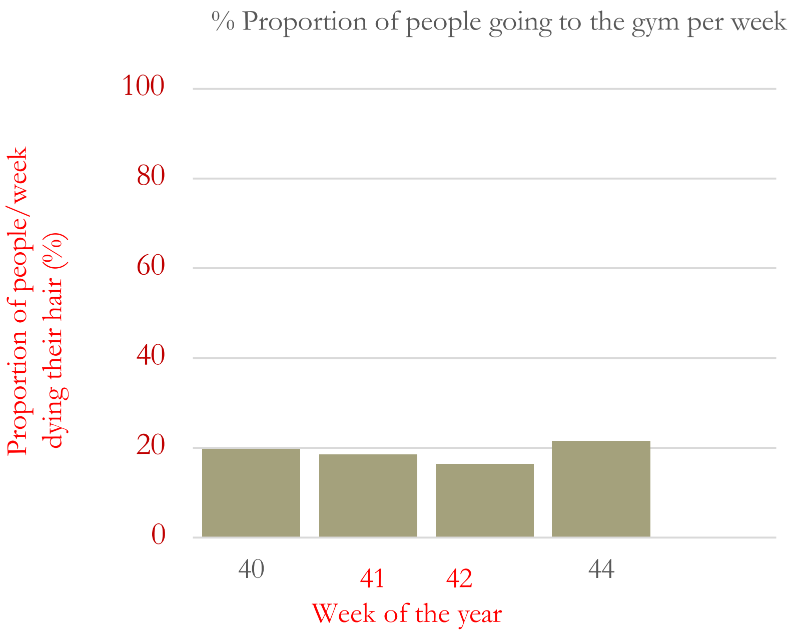
- colorblind – friendly color schemes

- combine colors with symbols, line types, and other design elements
- **use colors carefully**

https://blog.datawrapper.de/colorblindness-part2/

Axes - guide or mislead





A correct scientific plot MUST have

- unambiguous, clear plot title, axis title
- axes with units
- axis showing proportions → from 0 to 100 % !!
- homogenous scales

Captions

- detailed
- fully explain everything in the figure
- standalone
- explain any geometries used

QUESTIONS?

© Graphics: wur.nl;